

CESE323 ERC1.0

Cloud Security Envisioning & Strategy

Student Lab Exercises Guide with Hints

Student Edition | Guided labs, evidence capture, reflection prompts, and non-solution hints

Course code	CESE323
Version	ERC1.0
Guide type	Student Lab Exercises Guide with Hints
Primary platform	Microsoft Azure, Microsoft Defender for Cloud, Microsoft Defender portal, Microsoft Entra ID
Recommended delivery	One-day instructor-led workshop with optional self-paced completion
Lab model	Instructor-selected core labs plus two workload protection tracks
Student note: This guide provides hints and verification checkpoints, not a full answer key. Your instructor may adapt the workload labs based on the available Azure subscription, permissions, budget approvals, and selected customer scenario.	

Document Control

Field	Value
Document title	CESE323 Student Lab Exercises Guide with Hints
Course title	Cloud Security Envisioning & Strategy
Edition	Student Edition
Release	ERC1.0
Prepared for	Skunkworks Academy courseware development
Intended audience	Students, security engineers, administrators, cloud operations teams, and workshop delegates
Assumed environment	Azure subscription with Microsoft Defender for Cloud enabled or available for enablement
Production safety posture	Read-only by default unless the instructor confirms written approval for live changes

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1. Course and Lab Design Basis

This lab guide is aligned to the CESE323 workshop goals of strategic alignment, unified visibility, threat reduction, workload hardening, and operational readiness. The lab flow moves from security strategy into practical Defender for Cloud configuration, CSPM analysis, workload protection, remediation planning, and live-environment review.

1.1 Student Lab Outcomes

- Validate access to the Azure portal, Microsoft Defender for Cloud, Microsoft Defender portal, Microsoft Entra admin center, and the course resource location.
- Create a lightweight cloud security risk register and map business priorities to security outcomes.
- Review Microsoft Defender for Cloud posture views including Secure Score, recommendations, regulatory compliance, asset inventory, and coverage workbooks.
- Identify misconfigurations, over-privileged identities, exposure paths, and workload protection gaps.
- Complete two selected workload protection labs from Storage, Containers, App Service, and AI Services / Microsoft Foundry.
- Prepare a remediation plan with evidence, ownership, priority, and operational next steps.
- Perform approved cleanup and submit a structured lab evidence pack.

1.2 Lab Selection Model

Lab group	Required?	Purpose
Readiness and safety	Required	Ensure access, MFA, permissions, cost controls, and data handling rules are in place.
Strategy and governance	Required	Translate business context into cloud security objectives and risk priorities.
Defender for Cloud posture	Required	Build unified visibility across posture, recommendations, compliance, and coverage.
Workload protection	Select 2 of 4	Harden selected critical workload types based on the instructor-selected scenario.
Threat mapping and remediation	Required	Convert findings into prioritized remediation actions and executive-ready reporting.
Cleanup	Required	Reduce cost, remove temporary access, and leave the environment safe.
Important: Your instructor will confirm which two workload protection labs are mandatory for your delivery. Complete only the workload labs that match the approved environment and budget.		

2. Student Lab Rules and Safety Requirements

2.1 Safety Rules

- Use a lab subscription or assigned lab resource group unless the instructor explicitly authorizes a live customer review.
- Do not upload customer data, personal information, HR data, legal records, financial records, source code, secrets, credentials, or proprietary AI prompts.
- Do not enable paid Defender plans outside the approved scope.
- Do not assign yourself or others broad privileges unless the lab explicitly requires it and the instructor approves it.
- Capture screenshots and notes, but redact subscription IDs, tenant IDs, user IDs, customer names, IP addresses, secrets, and sensitive resource names before external sharing.
- Treat all generated alerts, recommendations, and logs as potentially sensitive operational information.

2.2 Required Evidence Standard

Evidence type	Minimum requirement
Screenshot	Use clear screenshots showing the relevant blade or result. Avoid exposing sensitive identifiers.
Observation note	Summarize what you observed and why it matters to the security posture.
Risk statement	Describe the asset, threat, weakness, impact, likelihood, and suggested control.
Remediation proposal	Include owner, priority, effort, dependency, and rollback consideration.
Reflection answer	Answer the lab reflection questions in your own words.

2.3 Naming Standard

Use this naming convention where you create resources:

rg-cese323-lab-<studentname>
stcese323<uniqueid>
app-cese323-<studentname>
aks-cese323-<studentname>
ai-cese323-<studentname>

Hint: Use short lowercase names where Azure naming rules require them. Storage account names must be globally unique and cannot contain hyphens.

3. Lab Map

Lab	Title	Track	Estimated time	Core output
0	Access, Safety, and Lab Readiness Check	Core	20 min	Readiness confirmation and evidence folder
1	Strategic Risk and Governance Mapping	Core	35 min	Risk register and security objectives
2	Microsoft Defender for Cloud Orientation and Coverage Baseline	Core	40 min	Coverage and baseline posture notes
3	CSPM Analysis: Secure Score, Recommendations, and Attack Paths	Core	55 min	Prioritized posture findings
4	Workload Protection Plan Selection	Core	30 min	Decision matrix for two workload labs
5	Defender for Storage	Optional workload	45 min	Storage protection evidence and risk notes
6	Defender for App Service	Optional workload	45 min	App Service protection evidence and risk notes
7	Defender for Containers	Optional workload	60 min	AKS/container protection evidence and risk notes
8	Defender for AI Services and Microsoft Foundry	Optional workload	60 min	AI security posture evidence and data-safety notes
9	Threat Mapping and Remediation Planning	Core	45 min	Remediation backlog and threat map
10	Operational Review and Executive Summary	Core	40 min	Operational review summary
11	Cleanup, Cost Control, and Evidence Submission	Core	30 min	Cleanup checklist and final evidence pack

Lab 0: Access, Safety, and Lab Readiness Check

Estimated time	20 minutes
Delivery mode	Individual or pair-based
Lab track	Core
Student output	Completed readiness checklist and initial evidence folder

Scenario

You are joining a cloud security strategy workshop that includes live portal review and optional configuration of Microsoft Defender for Cloud plans. Before touching the environment, you must verify access, permissions, and safety controls.

Task 1: Confirm identity and tenant access

1. Sign in with your assigned named account.
2. Confirm MFA is configured and working.
3. Open the Azure portal and confirm the correct directory / tenant is selected.
4. Record the tenant display name and lab subscription display name in your evidence notes. Do not record full tenant IDs unless your instructor asks for them.

Hint: Check the account selector in the top-right of the Azure portal. If resources are missing, you may be in the wrong directory rather than missing permissions.

Evidence to capture: Screenshot showing the Azure portal home or subscription page with sensitive identifiers masked.

Task 2: Confirm Defender portal access

5. Open the Microsoft Defender portal.
6. Confirm you can see the navigation areas required by the instructor, such as Incidents, Alerts, Exposure Management, or Cloud Security.
7. Do not change any settings during this task.

Hint: Access may differ by license and role. If a blade is unavailable, note the exact access message and continue with the Azure portal tasks.

Evidence to capture: Screenshot or note confirming portal access status.

Task 3: Create your evidence workspace

8. Create a local folder named CESE323-Lab-Evidence-<yourname>.
9. Inside it create subfolders: screenshots, notes, exports, cleanup.
10. Create a notes file named lab-notes.md or lab-notes.docx.

Hint: Evidence organization is part of the assessment discipline. Keep filenames meaningful, for example Lab03-SecureScore-Baseline.png.

Evidence to capture: Screenshot or file listing of your evidence folder structure.

Reflection Questions

- Why should each student use a named account instead of a shared administrator account?
- What could go wrong if a paid Defender plan is enabled at the wrong scope?
- Which access issue would block you from completing hands-on configuration labs?

Lab 1: Strategic Risk and Governance Mapping

Estimated time	35 minutes
Delivery mode	Small group or individual
Lab track	Core
Student output	Cloud security objectives, risk register, and prioritization notes

Scenario

Your organization wants to improve cloud security posture while aligning remediation effort to business risk. You must translate business priorities into measurable security outcomes before selecting technical controls.

Task 1: Define business-critical assets

11. List three business-critical cloud assets or services in scope for the workshop.
12. For each asset, identify the business owner, technical owner, data sensitivity, and operational criticality.
13. Mark whether the asset is production, pre-production, training, or demo.

Hint: A useful asset description includes what the asset does for the business, not only its Azure resource type.

Evidence to capture: Completed asset table in your notes.

Task 2: Create a risk register

14. For each asset, describe one plausible cloud security risk.
15. Identify the weakness or exposure, such as public access, weak identity control, missing logging, unprotected workload, or unclear ownership.
16. Assign preliminary likelihood and impact as Low, Medium, or High.

Hint: Use the wording: Because <condition>, <threat actor or failure mode> could <event>, causing <business impact>.

Evidence to capture: Risk register with at least three risks.

Task 3: Map risks to security goals

17. Group risks into objectives such as reduce exposed attack surface, strengthen identity control, improve workload protection, improve detection, or improve compliance evidence.
18. Choose the top two objectives for the remainder of the labs.
19. Define one success metric for each objective.

Hint: A good success metric is observable in Defender for Cloud, for example fewer high-severity recommendations, better Secure Score, or workload protection coverage enabled for selected resources.

Evidence to capture: Two prioritized security objectives with success metrics.

Student Worksheet: Risk Register

Asset	Business impact	Observed or suspected risk	Likelihood	Impact	Control idea	Owner

Lab 2: Microsoft Defender for Cloud Orientation and Coverage Baseline

Estimated time	40 minutes
Delivery mode	Instructor-led followed by individual evidence capture
Lab track	Core
Student output	Baseline observations of Defender for Cloud coverage, Secure Score, and enabled plans

Scenario

You need a baseline view of the environment before recommending workload protection changes. This lab helps you locate the core Microsoft Defender for Cloud areas used throughout the course.

Task 1: Open Microsoft Defender for Cloud

20. In the Azure portal, search for Microsoft Defender for Cloud.
21. Open Overview and record the key posture indicators that are visible to your role.
22. Open Environment settings and select the approved lab subscription.

Hint: If Environment settings is not visible, you may have Reader-only access or a role that lacks security management permissions. Ask the instructor before requesting more privileges.

Evidence to capture: Screenshot of Defender for Cloud Overview and Environment settings with sensitive identifiers masked.

Task 2: Review current coverage

23. Locate the area that shows Defender plan coverage or use the Coverage workbook if available.
24. Record which plans are On, Off, Trial, or Not available.
25. Compare the enabled plans to the labs selected for this course.

Hint: Coverage is not the same as security quality. A plan can be enabled while resources still have unresolved recommendations.

Evidence to capture: Coverage note listing selected plan states.

Task 3: Review posture entry points

26. Open Recommendations.
27. Open Secure Score or security posture score view if available.
28. Open Regulatory compliance if available.
29. Open Workbooks or dashboards if available.

Hint: Take notes on what you can see, what is empty, and what requires additional permissions or data collection time.

Evidence to capture: At least two screenshots showing posture entry points.

Reflection Questions

- What is the difference between enabling a Defender plan and remediating recommendations?
- Why is a baseline important before making any configuration change?
- Which Defender for Cloud area gave you the clearest view of the environment?

Lab 3: CSPM Analysis: Secure Score, Recommendations, and Attack Paths

Estimated time	55 minutes
Delivery mode	Individual or pair-based
Lab track	Core
Student output	Prioritized posture findings and attack path notes

Scenario

Your security team needs to reduce the highest-risk configuration weaknesses. You will review CSPM signals, prioritize findings, and identify which recommendations should become remediation actions.

Task 1: Capture the posture baseline

30. Open Secure Score or the equivalent posture score view.
31. Record the score or visible posture indicators.
32. Identify the top recommendation categories shown in your environment.

Hint: Do not treat the score as the only decision metric. Tie score changes to business risk and critical workloads.

Evidence to capture: Baseline score or posture note with date, time, subscription, and visible categories.

Task 2: Analyze recommendations

33. Open Recommendations.
34. Filter by severity, resource type, or affected resource count.
35. Select three recommendations that appear relevant to the course objectives.
36. For each recommendation, record affected resource type, severity, remediation guidance summary, and possible business impact.

Hint: Prioritize recommendations that affect internet-facing resources, identity boundaries, logging, encryption, or workload protection coverage.

Evidence to capture: Three recommendation summaries in your evidence notes.

Task 3: Review attack paths or exposure paths

37. Open the attack path, exposure management, or cloud security graph view if available.
38. Identify whether any path links a vulnerable resource to a critical asset.
39. Document what the path suggests without changing the environment.

Hint: If no attack path is shown, note why: no data, insufficient permissions, no critical assets identified, or feature not enabled in the lab scope.

Evidence to capture: Screenshot or observation note for the attack path / exposure view.

Task 4: Create a posture finding summary

40. Choose the top three findings from this lab.
41. Rank them P1, P2, and P3 based on risk, effort, business criticality, and dependency.
42. Write one recommended next action for each finding.

Hint: Use this phrasing: Prioritize <finding> because it affects <asset/scope> and reduces <risk/outcome>.

Evidence to capture: Posture finding summary table.

Student Worksheet: Posture Findings

Priority	Finding	Affected resource / scope	Security impact	Suggested next action	Owner
P1					
P2					
P3					

Lab 4: Workload Protection Plan Selection

Estimated time	30 minutes
Delivery mode	Group decision exercise
Lab track	Core
Student output	Decision matrix selecting two workload protection labs

Scenario

The workshop design allows only two live workload protection tracks. Your group must select the two modules that create the highest value while respecting cost, permissions, and available lab resources.

Task 1: Review workload candidates

43. Review the four workload lab options: Storage, Containers, App Service, and AI Services / Microsoft Foundry.
44. For each option, confirm whether the required lab resource exists or can be created safely.
45. Confirm whether enabling the relevant plan is approved.

Hint: A valuable workload lab is one where students can observe a meaningful security signal, not only toggle a plan to On.

Evidence to capture: Candidate list with Ready, Not Ready, or Demo Only status.

Task 2: Complete the workload decision matrix

46. Score each workload from 1 to 5 for business relevance, lab readiness, risk reduction, cost control, and learning value.
47. Select the two highest-value labs.
48. Write a short justification for each selected workload.

Hint: If there is a tie, choose the workload with safer lab isolation and lower production risk.

Evidence to capture: Completed decision matrix and selected workload tracks.

Student Worksheet: Workload Selection Matrix

Workload	Business relevance	Lab readiness	Risk reduction	Cost control	Learning value	Selected?
Defender for Storage						
Defender for Containers						
Defender for App Service						
Defender for						

Workload	Business relevance	Lab readiness	Risk reduction	Cost control	Learning value	Selected?
AI Services / Foundry						

Lab 5: Defender for Storage

Estimated time	45 minutes
Delivery mode	Optional workload lab - complete only if assigned
Lab track	Workload option A
Student output	Storage protection configuration evidence and risk notes

Scenario

A business unit stores sensitive operational files in Azure Storage. You need to review whether storage workloads have appropriate threat protection and safe handling controls for uploads and access patterns.

Safety: Use test files only. Do not upload real customer data, malware, credentials, personal information, or confidential documents.

Task 1: Identify or create the lab storage scope

49. Confirm the instructor-approved resource group.
50. Identify an existing lab storage account or create one only if instructed.
51. Confirm that the storage account belongs to the approved lab subscription or resource group.

Hint: Storage accounts have strict naming rules. If creation fails, check global uniqueness, lowercase characters, and region availability.

Evidence to capture: Storage account overview screenshot with sensitive identifiers masked.

Task 2: Review Defender for Storage plan status

52. In Defender for Cloud, open Environment settings.
53. Select the approved subscription or resource scope.
54. Locate the Storage plan and record whether it is enabled.
55. If enabled, review available settings. Do not enable optional charged capabilities unless approved by the instructor.

Hint: Some features are subscription-level and some can be resource-level depending on the plan and configuration model. Read the scope carefully before saving changes.

Evidence to capture: Plan status screenshot or written note.

Task 3: Review storage security recommendations

56. Open Recommendations and filter for storage-related findings.
57. Record two relevant recommendations or state that none are currently visible.
58. For each recommendation, identify the control category: access, encryption, logging, network exposure, malware scanning, or data discovery.

Hint: Recommendations may take time to appear after resources are created or after plans are enabled. Note data

freshness in your evidence.

Evidence to capture: Two storage recommendation summaries or a no-findings observation.

Task 4: Assess upload and access risk

59. Upload only an instructor-approved harmless test file if instructed.
60. Review whether the configured protection would help detect suspicious uploads or access patterns.
61. Record a risk statement and a mitigation recommendation.

Hint: The goal is not to trigger malware detection. The goal is to understand where protection is configured and what evidence would be available.

Evidence to capture: Risk statement and mitigation note.

Reflection Questions

- What storage risks can Defender for Storage help detect or reduce?
- Why is it important to define plan scope before enabling storage protection?
- What evidence would you present to a business owner to justify storage protection?

Lab 6: Defender for App Service

Estimated time	45 minutes
Delivery mode	Optional workload lab - complete only if assigned
Lab track	Workload option B
Student output	App Service protection evidence and application risk notes

Scenario

A team hosts a customer-facing web application on Azure App Service. You need to evaluate App Service protection, posture recommendations, and runtime monitoring considerations without disrupting the application.

Task 1: Identify the app workload

62. Open the approved lab App Service or sample application.
63. Record the app service plan, runtime stack, region, and environment classification.
64. Confirm that the app is non-production or approved for read-only review.

Hint: If there is no sample app, the instructor may demonstrate this lab rather than requiring live configuration.

Evidence to capture: App Service overview screenshot with non-sensitive details.

Task 2: Review Defender for App Service plan status

65. Open Defender for Cloud > Environment settings.
66. Select the approved subscription.
67. Locate the App Service plan row and review status.
68. If your role allows it and the instructor approves, enable the plan for the approved scope.

Hint: Do not confuse App Service Plan as a compute resource with the Defender for App Service security plan. They are related but not the same thing.

Evidence to capture: Plan status screenshot or written note.

Task 3: Review application recommendations

69. Filter Defender for Cloud recommendations for App Service or web application findings.
70. Identify at least two relevant security posture issues where available.
71. Classify each issue as platform configuration, identity/access, network exposure, logging, TLS, or runtime monitoring.

Hint: Look for quick wins that reduce exposure without requiring application code changes.

Evidence to capture: Application recommendations summary.

Task 4: Create an application protection note

72. Write a concise recommendation for how the organization should monitor and protect this application.
73. Include one preventive control, one detective control, and one operational owner.
74. State one risk that remains outside Defender for App Service and may require SDLC or code-level controls.

Hint: Think in layers: platform configuration, identity, network controls, app code, logging, incident response.

Evidence to capture: Application protection note in your lab evidence.

Reflection Questions

- Which application risks are visible from cloud posture tools and which require developer input?
- What is the operational value of enabling runtime threat detection for App Service?
- How would you explain App Service security posture to a non-technical application owner?

Lab 7: Defender for Containers

Estimated time	60 minutes
Delivery mode	Optional workload lab - complete only if assigned
Lab track	Workload option C
Student output	Container / AKS protection evidence and risk notes

Scenario

A platform team operates containerized workloads on Kubernetes. You need to review container security coverage, vulnerability signals, and runtime protection without destabilizing the cluster.

Cost and safety: AKS clusters can generate cost even when idle. Do not create or scale clusters unless your instructor explicitly assigns that task.

Task 1: Validate cluster scope

75. Identify the approved AKS cluster or container environment.
76. Record the cluster name, resource group, region, and environment classification.
77. Confirm that you have permission to view the cluster and its security recommendations.

Hint: If the cluster is unavailable, complete this lab using the instructor demonstration and record observations rather than creating a new cluster.

Evidence to capture: Cluster overview screenshot or demo observation note.

Task 2: Review Defender for Containers plan status

78. Open Defender for Cloud > Environment settings.
79. Select the subscription where the cluster resides.
80. Locate Containers and review whether it is On.
81. If assigned, review component settings such as sensors, registry access, Kubernetes API access, agentless discovery, or vulnerability assessment.

Hint: Component names and availability may vary by cloud, cluster type, plan state, and current Microsoft feature updates. Capture what your environment shows.

Evidence to capture: Containers plan status and component settings screenshot.

Task 3: Analyze container security recommendations

82. Open Recommendations and filter for Kubernetes, AKS, container, registry, or image findings.
83. Identify at least two findings where available.
84. For each finding, classify whether it relates to image vulnerability, cluster configuration, identity/RBAC, network exposure, or runtime protection.

Hint: Container risks often combine multiple layers: image, registry, cluster control plane, workload identity, network policy, and runtime behavior.

Evidence to capture: Container recommendation summary table.

Task 4: Develop a container hardening action

85. Select one container-related finding.
86. Write a remediation action that includes dependency, expected impact, validation method, and rollback consideration.
87. Do not apply remediation unless the instructor approves it.

Hint: Good remediation plans include how you will prove the risk was reduced after the change.

Evidence to capture: One container hardening action plan.

Reflection Questions

- Why is container security more than scanning container images?
- What is the difference between posture management and runtime threat detection for containers?
- Which cluster security finding would you prioritize first and why?

Lab 8: Defender for AI Services and Microsoft Foundry

Estimated time	60 minutes
Delivery mode	Optional workload lab - complete only if assigned
Lab track	Workload option D
Student output	AI security posture evidence, data safety notes, and AI workload risk mapping

Scenario

Your organization is adopting AI services and Microsoft Foundry workloads. You need to evaluate AI security posture, identity boundaries, data leakage risk, prompt-related monitoring, and Defender for Cloud coverage.

AI data handling rule: Never enter confidential, personal, customer, HR, financial, legal, credential, source code, or proprietary data into AI prompts during this lab.

Task 1: Identify the approved AI scope

88. Confirm whether the lab uses Azure AI services, Azure OpenAI, Microsoft Foundry, or an instructor demonstration.
89. Record the approved AI resource group and service type.
90. Confirm that the environment is sandbox, demo, or explicitly approved for review.

Hint: AI labs are often limited by regional availability, licensing, model access, and governance approval. Document constraints rather than bypassing them.

Evidence to capture: AI scope note and screenshot where permitted.

Task 2: Review AI protection plan status

91. Open Defender for Cloud > Environment settings.
92. Select the relevant subscription.
93. Locate the AI Services or AI workload protection plan and review status.
94. If instructed, review settings related to data security for AI interactions or prompt evidence handling.

Hint: Prompt evidence can be sensitive. Understand masking, storage, access, and approval implications before changing settings.

Evidence to capture: AI plan status screenshot or demo observation note.

Task 3: Review Data and AI security posture

95. Open the relevant Data and AI security dashboard, AI posture view, or Defender recommendation view if available.
96. Identify any AI-related recommendations, alerts, or configuration findings.
97. Classify each observation as identity, network, data protection, model/prompt safety, logging, or governance.

Hint: If the dashboard has limited data, the environment may not yet have supported AI resources, protection may not be enabled, or telemetry may need time to populate.

Evidence to capture: AI security observation summary.

Task 4: Map AI workload risks

98. Write three risks specific to AI workloads.
99. Include at least one risk related to identity/access, one related to data leakage, and one related to prompt or model behavior.
100. For each risk, suggest one preventive and one detective control.

Hint: Use the shared responsibility model: platform security, cloud configuration, workload identity, data governance, user behavior, and model interaction controls all matter.

Evidence to capture: AI workload risk table.

Student Worksheet: AI Workload Risk Mapping

Risk area	Risk statement	Preventive control	Detective control	Owner
Identity / access				
Data leakage				
Prompt / model behavior				

Lab 9: Threat Mapping and Remediation Planning

Estimated time	45 minutes
Delivery mode	Individual or small group
Lab track	Core
Student output	Prioritized remediation backlog and threat map

Scenario

You have now collected posture and workload evidence. The next step is to translate findings into remediation actions that are traceable, risk-based, and operationally achievable.

Task 1: Select findings for remediation

101. Select five findings from Labs 2 to 8.
102. Include at least one posture finding and one workload protection finding.
103. Exclude findings that are out of scope or cannot be validated in the current lab.

Hint: Pick findings that matter to business risk, not only findings that are easy to screenshot.

Evidence to capture: List of five selected findings.

Task 2: Map each finding to threat or control category

104. For each finding, choose a category such as identity compromise, exposed resource, missing monitoring, vulnerable workload, data exposure, or compliance gap.
105. Add a business impact statement.
106. Add a recommended mitigation strategy.

Hint: You can use a practical mapping approach even if you do not know the exact ATT&CK technique. Start with the attack outcome: unauthorized access, persistence, exfiltration, lateral movement, or service disruption.

Evidence to capture: Threat mapping table.

Task 3: Build a remediation backlog

107. Assign each finding a priority: P1 immediate, P2 planned, P3 backlog, or Exception / accepted risk.
108. Estimate effort: Low, Medium, or High.
109. Assign an owner role: cloud platform, security operations, identity team, application team, data owner, or governance team.
110. Define validation evidence for completion.

Hint: Every remediation item should have a measurable closure condition. Avoid vague actions such as Improve security.

Evidence to capture: Remediation backlog table.

Student Worksheet: Remediation Backlog

Priority	Finding	Threat / control category	Owner role	Effort	Validation evidence	Status
P1						
P2						
P2						
P3						
Exception						

Lab 10: Operational Review and Executive Summary

Estimated time	40 minutes
Delivery mode	Group exercise
Lab track	Core
Student output	Operational review summary suitable for customer or leadership discussion

Scenario

You must present the lab findings in a format that helps decision makers approve next steps. Your summary should connect business objectives, posture evidence, workload risk, and immediate remediation actions.

Task 1: Prepare the operational review

111. Summarize the initial security objectives from Lab 1.

112. Summarize baseline coverage and posture observations from Labs 2 and 3.
113. Summarize workload lab observations from the two selected workload labs.
114. Identify blockers such as permissions, licensing, missing resources, feature availability, or data freshness.

Hint: Decision makers need to know what is known, what is not known, and what action is recommended next.

Evidence to capture: Operational review notes.

Task 2: Draft an executive summary

115. Write a one-page summary using the structure: Context, Key Observations, Top Risks, Recommended Actions, Decisions Required.
116. Limit technical jargon unless it directly supports a decision.
117. Include one statement on cost / scope control and one statement on production safety.

Hint: Use risk language, not tool language. For example: Unprotected storage can increase malware and data exposure risk rather than The Storage plan is Off.

Evidence to capture: One-page executive summary draft.

Task 3: Prepare talking points

118. Create three speaking points for a security leader.
119. Create three speaking points for a cloud operations owner.
120. Create three speaking points for an application or data owner.

Hint: Each stakeholder cares about different outcomes: risk reduction, operational effort, service reliability, compliance, budget, and accountability.

Evidence to capture: Stakeholder talking points.

Executive Summary Template

Section	Student notes
Context	
Key observations	
Top risks	
Recommended actions	
Decisions required	
Production safety / cost note	

Lab 11: Cleanup, Cost Control, and Evidence Submission

Estimated time	30 minutes
Delivery mode	Individual
Lab track	Core
Student output	Cleanup checklist and final evidence pack

Scenario

A good lab finishes by controlling cost, removing unnecessary access, documenting evidence, and leaving the environment safe. This is mandatory even if some labs were demonstration-only.

Task 1: Review resources created during labs

- 121. List all resources you created or modified.
- 122. Confirm whether each item should be retained, stopped, disabled, or deleted.
- 123. Do not delete shared resources unless the instructor confirms that they belong to your lab group.

Hint: Filter by resource group and tag. Check created time and owner notes before deleting anything.

Evidence to capture: Resource cleanup list.

Task 2: Confirm Defender plan cleanup

- 124. Review which Defender plans were enabled for lab purposes.
- 125. Confirm with the instructor whether plans must remain enabled or be disabled.
- 126. Record any plan that remains enabled and why.

Hint: Do not disable a plan in a customer environment unless explicitly instructed. Cleanup decisions must align with the approved scope.

Evidence to capture: Plan cleanup status note.

Task 3: Remove temporary access and review budget

- 127. Confirm whether any temporary role assignments were created.
- 128. Request removal of temporary student access where appropriate.
- 129. Check budget alerts or cost management views if you have access.

Hint: Student evidence should not include full billing data unless the instructor asks for it. Record only whether cost control was reviewed.

Evidence to capture: Access and cost review note.

Task 4: Submit the evidence pack

- 130. Collect screenshots, notes, worksheets, and executive summary into your evidence folder.
- 131. Redact sensitive identifiers before submission.
- 132. Submit using the instructor-approved LMS, Teams, GitHub, or shared folder process.

Hint: Before submitting, check that each lab has at least one observation, one evidence item, and one reflection answer.

Evidence to capture: Final evidence pack submitted.

Cleanup Checklist

Item	Action required	Student status
Defender paid plans	Retain only if approved; otherwise disable at approved scope.	
AKS clusters	Delete or scale down if lab-created and no longer needed.	
Storage accounts	Delete if lab-created after evidence is captured.	
App Services	Stop or delete if lab-created.	
AI resources / model deployments	Delete or scale down if lab-created and approved for cleanup.	
Role assignments	Remove temporary access assignments.	
Budget alerts / spend	Review and record status.	
Evidence	Redact and submit.	

Appendix A: Final Evidence Pack Structure

Folder / file	Required content
00-readiness	Access confirmation, tenant/subscription confirmation, MFA and portal access notes.
01-strategy	Risk register, security objectives, success metrics.
02-defender-baseline	Defender for Cloud overview, coverage, Secure Score, recommendations.
03-cspm-analysis	Three prioritized posture findings and any attack path / exposure observations.
04-workload-selection	Workload decision matrix and selected tracks.
05-workload-labs	Evidence from the two assigned workload labs.
06-remediation	Threat mapping and remediation backlog.
07-executive-summary	One-page operational review summary.
08-cleanup	Cleanup checklist and final status notes.

Appendix B: Troubleshooting Hints

Symptom	Likely cause	Student action
Subscription not visible	Wrong directory, missing role assignment, expired invitation, or account mismatch.	Switch directory, sign out/in, or ask instructor to validate assignment.
Environment settings missing	Insufficient permission or Defender for Cloud not available in visible scope.	Record access message; continue with read-only tasks until instructor confirms role.
Recommendations are empty	No supported resources, new resource telemetry not populated, plan not enabled, or insufficient permissions.	Capture observation and note data freshness; do not assume no risk exists.
Plan cannot be enabled	Permission, budget, policy, subscription type, or regional limitation.	Do not attempt privilege escalation; document blocker for Lab 10.
AI dashboard has no data	No supported AI resource, plan not enabled, telemetry delay, or unsupported scenario.	Complete AI risk mapping using approved demo observations.
Container findings unavailable	No AKS cluster, unsupported version, component disabled, or data not yet collected.	Use instructor demo and complete risk analysis rather than creating unsanctioned clusters.
Unexpected cost warning	Paid Defender plan or compute resource may be active.	Pause and notify instructor before proceeding.

Appendix C: Quick Glossary

Term	Meaning in this course
CSPM	Cloud Security Posture Management. Used to identify configuration risks, compliance gaps, and exposure paths.
CWPP	Cloud Workload Protection Platform. Used to protect workloads such as servers, containers, storage, databases, and AI services.
Secure Score	A posture indicator that helps prioritize security recommendations and track improvement.
Recommendation	A Defender for Cloud finding with suggested remediation guidance.
Attack path	A chain of weaknesses that could allow a threat actor to reach a critical asset.
Workload protection plan	A Defender for Cloud plan that enables deeper protection and detection for a workload category.
Least privilege	Grant only the permissions required to complete a task and no more.
Break-glass account	Emergency account reserved for recovery scenarios, not normal student lab work.
Evidence pack	The student submission containing screenshots, worksheets, observations, and cleanup proof.

Appendix D: References and Source Basis

The guide was prepared from the CESE323 Cloud Security Envisioning & Strategy workshop syllabus, the student pre-course readiness requirements, and current Microsoft Learn documentation reviewed during courseware generation.

- Microsoft Defender for Cloud overview: <https://learn.microsoft.com/en-us/azure/defender-for-cloud/defender-for-cloud-introduction>
- Connect Azure subscriptions to Microsoft Defender for Cloud: <https://learn.microsoft.com/en-us/azure/defender-for-cloud/connect-azure-subscription>
- User roles and permissions in Microsoft Defender for Cloud: <https://learn.microsoft.com/en-us/azure/defender-for-cloud/permissions>
- Prerequisites for Microsoft Defender for Storage: <https://learn.microsoft.com/en-us/azure/defender-for-cloud/support-matrix-defender-for-storage>
- Enable Defender for Containers in Microsoft Defender for Cloud: <https://learn.microsoft.com/en-us/azure/defender-for-cloud/defender-for-containers-enable-plan>
- Protect applications with Microsoft Defender for App Service: <https://learn.microsoft.com/en-us/azure/defender-for-cloud/tutorial-enable-app-service-plan>
- Enable threat protection for AI services: <https://learn.microsoft.com/en-us/azure/defender-for-cloud/ai-onboarding>
- Azure Monitor workbooks with Defender for Cloud data: <https://learn.microsoft.com/en-us/azure/defender-for-cloud/custom-dashboards-azure-workbooks>